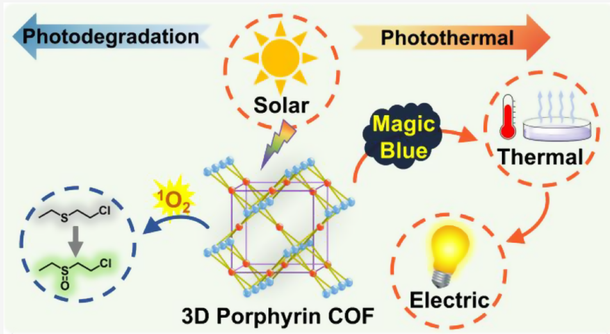
JACS
JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
pubs.acs.org/JACS
Article

Exploring Three-Dimensional Porphyrin-Based Covalent Organic Frameworks with Outstanding Solar Energy Conversion

Jiaqi Guo,○ Yunyu Guo,○ Mingxuan Zhang, Hongde Yu, Haohao Bi, Jingyi Feng, Jiahuan Wang, Keyu Geng, Thomas Heine, Junliang Sun, Di Li,* Donglin Jiang,* and Enquan Jin*

Cite This: J. Am. Chem. Soc. 2025, 147, 30369−30379

ABSTRACT: As an emerging class of porous aromatic polymers, porphyrin-based covalent organic frameworks (COFs) have been widely employed in assorted applications due to their unique electronic configurations and properties. Notably, three-dimensional (3D) COFs, characterized by porphyrin cores exposed along steric ordered nanochannels, exhibit great promise for solar energy conversion. However, the development of 3D porphyrin-based COFs continues to present a synthetic challenge, stemming from the scarcity of appropriate topotactic designs and appropriate building blocks. In this study, a series of 3D Por-An-COFs with a reasonable 2-fold lvt-b topology have been synthesized. The 3D architecture enables the periodic alignment of porphyrin units within the conjugated backbones, facilitating light harvesting and interactions between guest species and active centers. Consequently, the 3D Por-An-COF features superior photoresponsive characteristics, including high solar-to-chemical and solar-to-thermal conversion capabilities. In particular, the interfacial water evaporation system based on 3D Por-An-COF achieved an evaporation rate of 1.64 kg m^{-2} h^{-1}, and the related thermoelectric device generates an output voltage of 195 mV. This research not only extends the structural diversity of 3D porphyrin-based COFs for photoenergy conversion but also elucidates the intrinsic dimensionality-dependent photoresponsive behaviors, providing valuable insights for the development of advanced porphyrin-based photosensitizers for a wide range of applications.



■ INTRODUCTION

The exploration of remarkable photoresponsive materials capable of harnessing solar energy for both exceptional mass and energy conversion has become a focal point in the sustainable development of society.[1,2] With state-of-the-art efforts, covalent organic frameworks (COFs) have emerged as promising candidates due to their tunable structures and unique electronic properties.[3−15] Specifically, porphyrin-based COFs, which incorporate porphyrin units into ordered and porous lattices, have garnered broad attention for their capability to integrate multiple solar energy-related functionalities,[16−19] including photodegradation and photothermal conversion.

Photodegradation, a green procedure that employs solar energy to break down organic pollutants such as dyes and chemical warfare agents, provides a pivotal pathway to environmental remediation.[20,21] The efficiency of this process is substantially dependent on the capacity of materials to harvest light and generate reactive species, such as radicals and excited states, which facilitate the decomposition of contaminants.[22] Recent advancements in porphyrin-based COFs have highlighted their potential in the realm of photodegradation, owing to their tailorable and intense tendency to capture light energy for conducting the corresponding chemical reactions.[23−28]

In addition, photothermal conversion, another primary functionality of photoresponsive materials, possesses broad implications in various domains,[29−32] including water purification,[30] sterilization,[31] and cancer therapy.[32] The performance of photothermal conversion hinges on the propensity to absorb visible light across a wide spectral range and subsequently convert it into thermal energy with high efficiency.[33−35] Thereby, porphyrin-based COFs are particularly well-suited for these applications, attributed to their unique extended π-conjugation system and the narrow bandgap of porphyrin units within the aromatic networks.

To date, porphyrin-based COFs endowed with the combined capabilities of pollutant remediation and considerable thermal energy generation have been scarcely reported

Received: June 11, 2025
Revised: August 2, 2025
Accepted: August 4, 2025
Published: August 12, 2025

https://doi.org/10.1021/jacs.5c09844
J. Am. Chem. Soc. 2025, 147, 30369−30379

due to the lack of suitable topologies and functional building blocks. This dual functionality is desirable, as the thermal energy produced during photothermal conversion may boost the efficacy of photodegradation. The heat raises the kinetic energy of the molecules, thereby accelerating the degradation process of pollutants.[36] Conversely, the reactive species yielded during photodegradation potentially contribute to the overall thermal output, improving the photothermal conversion efficiency.[37]

Furthermore, the majority of porphyrin-based COFs reported in the literature are characterized by two-dimensional (2D) layered stacking structures or nonfully conjugated three-dimensional (3D) networks,[38−41] which often result in the encapsulation of a significant portion of photoresponsive porphyrin cores within densely packed lattices or confine the rapid charge carrier transport. Therefore, the exploration of photoresponsive 3D porphyrin-based COFs with all conjugated backbones has been a heavily pursued goal, holding substantial promise in offering comprehensive solutions to environmental and energy-related challenges.

This study delves into the exploration of 3D porphyrin-based Por-An-COFs, molecularly engineered with an unpresented 2-fold lvt-b topology, for both photodegradation and photothermal conversion. Through a series of benchmark experiments, the dimensionality-dependent photoresponsive characteristics of these COFs were systematically evaluated. Compared to their 2D counterparts, the 3D Por-An-COFs, constructed with sterically arranged porphyrin units, amplify the accessibility of guest molecules and light-harvesting efficiency (Figure 1a,b). As a result, these 3D COFs

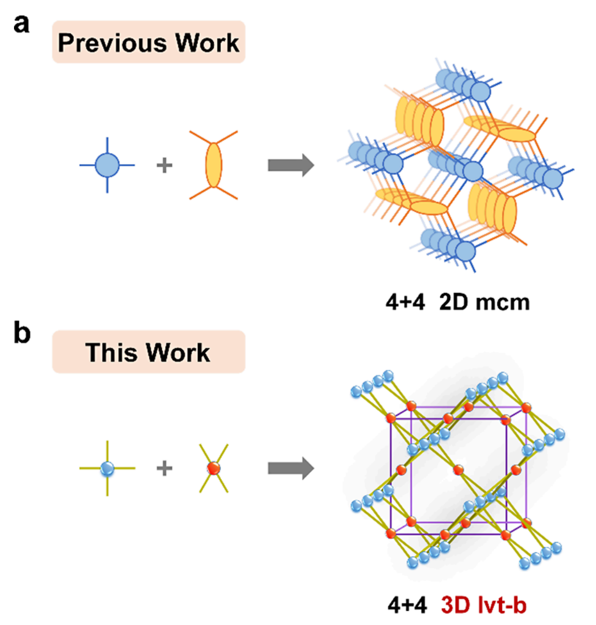


Figure 1. (a) Conventional 2D porphyrin-based COF with a 4 + 4 mcm topology. (b) 3D porphyrin-based COF in this work featuring a 4 + 4 lvt-b topology.

demonstrate enhanced performance relative to conventional porphyrin-based materials. Besides, the facile preparation and purification procedures of the robust 3D Por-An-COFs potentially facilitate their scalability and widespread adoption across various domains. This research not only elucidates the intrinsic relationship between dimensionality and efficient solar energy conversion but also provides a scaffold for the development of multifunctional photosensitive materials for sustainable chemistry and environmental remediation.

RESULTS AND DISCUSSION

Structural Design and Synthesis. In general, aligning porphyrin cores to the 3D lattice facilitates rapid molecular diffusion through steric nanochannels and enhances the interaction between guest molecules and exposed functional sites. Besides, the intended 3D porphyrin-based frameworks must ensure an efficient light harvesting nature by forming overall conjugated backbones. However, most reported 3D porphyrin-based COFs are not all conjugated skeletons. In addition, the polycondensation of well-known building blocks of COFs, e.g., D_{4h}-symmetric 5,10,15,20-tetra(4-aminophenyl)porphyrin (H_2P) and C_{2h}-symmetric monomers, commonly yields 2D COFs with mcm topology (Figure 1a). Herein, we explored a nonplanar C_2-symmetric unit, i.e., 4,4′,4″,4‴-(anthracene-9,10-diylidenebis(methanediylylidene))tetrabenzaldehyde (An) for the synthesis of 3D imine-linked porphyrin-based COFs with a reasonable lvt-b topology (Figure 1b).

The synthetic process involved the condensation of two 4-connected monomers: D_{4h}-symmetric nickel-coordinated 5,10,15,20-tetra(4-aminophenyl)porphyrin (NiP) and C_2-symmetric An, which served as linkers to form 3D porphyrin-based COF (NiP-An-COF). To achieve high crystallinity, we systematically screened the solvothermal conditions, including the selection of solvent, catalyst, and reaction temperature. The optimal conditions resulted in an 88% isolated yield of 3D NiP-An-COF via polymerization of NiP and An in 1,4-dioxane, with an acetic acid solution (6 M) as catalyst at 100 °C for 72 h (Figure 2a).

Powder X-ray diffraction (PXRD) analysis of the 3D NiP-An-COF revealed characteristic diffraction peaks at 3.33°, 5.95°, and 6.71°, corresponding to the (200), (002), and (202) reflection facets, respectively (Figure 2b, green curve). Pawley refinement based on the *Iba*2 space group with lvt-b topology provided optimized lattice parameters: a = 54.4991 Å, b = 29.5005 Å, c = 29.7506 Å, and $\alpha = \beta = \gamma = 90°$ (Table S1). The refined PXRD pattern closely matched the experimental data, with R_p = 5.16% and R_{wp} = 3.52% for 3D NiP-An-COF (Figure 2b, purple curve). The refined PXRD pattern of 3D NiP-An-COF reproduced the experimental result with minimal deviation (Figure 2b, black curve). These results indicate that the product adopts a 3D skeleton with 2-fold interwoven backbones, forming ordered channels with a pore size of 2.1 nm, as observed from both top and side views of the crystalline lattice (Figure 2c,d). Other topologies constructed with the same units did not match the experimentally observed PXRD patterns (Figure S1). Additionally, the crystalline structure of NiP-An-COF was further verified through continuous rotation electron diffraction (cRED) based on its clear diffraction signals. All cRED data were reconstructed and visualized with the program REDp.[42] Considering the symmetry of the building blocks, the 3D reciprocal lattice of NiP-An-COF was determined to be a *C*-centered orthorhombic Bravais lattice, with unit cell parameters of a = 52.65 Å, b = 30.18 Å, and c = 29.19 Å (Figure 2e). This structural model reveals a topology characterized as lvt-b with 2-fold interpenetration (Figure S2). High-resolution transmission electron microscopy (HR-TEM) images confirmed the presence of highly ordered porous architectures. The spacing between adjacent lattice fringes was approximately 2.1 nm, consistent with the

https://doi.org/10.1021/jacs.5c09844
J. Am. Chem. Soc. 2025, 147, 30369−30379

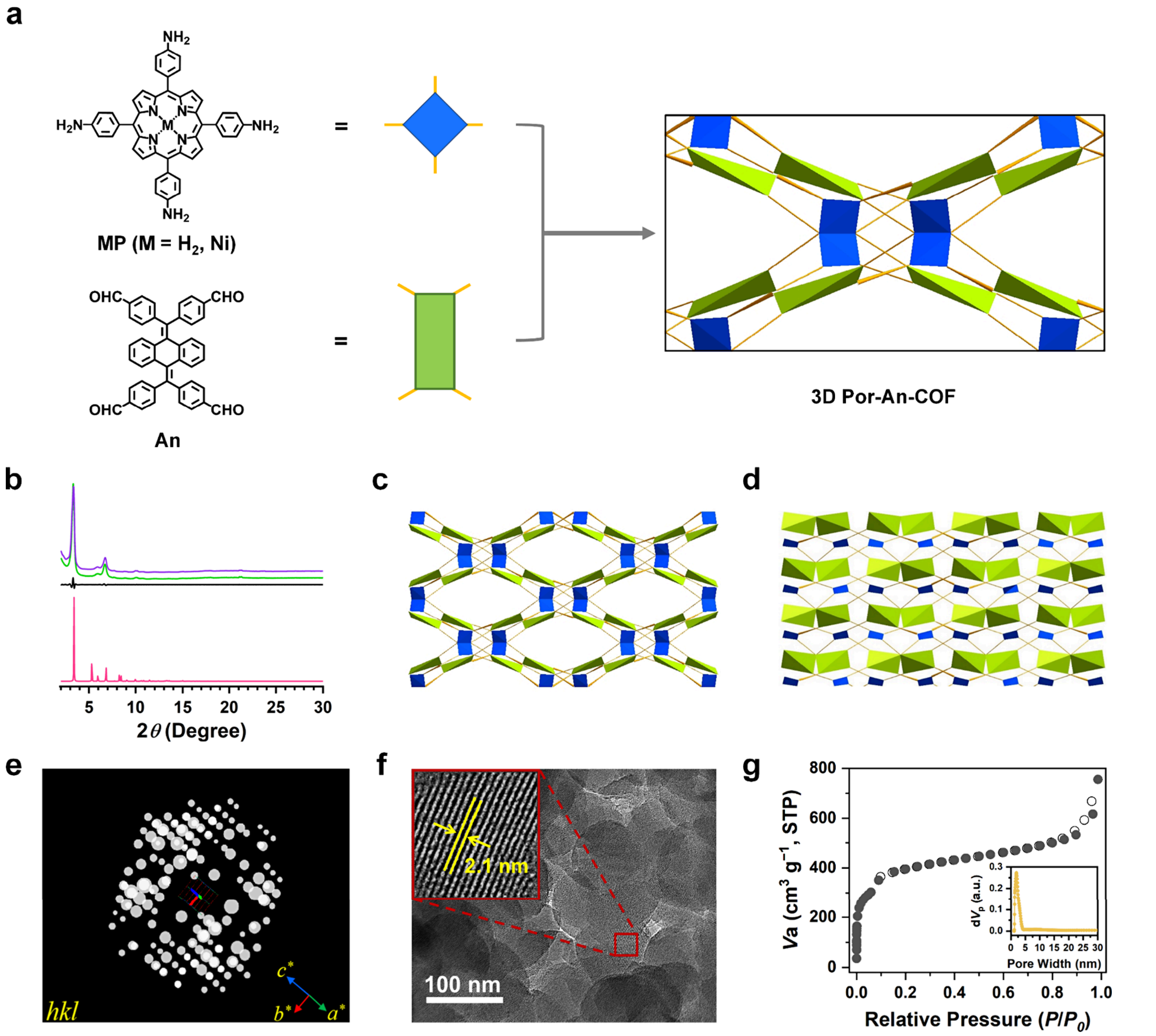


Figure 2. Design and characterization of crystal structures. (a) Synthesis of 3D Por-An-COF through the condensation of MP and An. (b) PXRD profiles of 3D NiP-An-COF. Experimentally observed (green curve), Pawley refined (purple curve), their difference (black curve), and simulated "lvt-b" topology structure (red curve). (c, d) Lattice structure viewed from the top and side. (e) cRED image of NiP-An-COF. (f) HR-TEM image of 3D NiP-An-COF. (g) Nitrogen sorption isotherm curves of 3D NiP-An-COF (solid dots for adsorption and open circles for desorption). Inset: pore-size distribution profile of 3D NiP-An-COF.

structural model (Figure 2f). The Brunauer−Emmett−Teller (BET) surface area of 3D NiP-An-COF was calculated to be 1302 m^2 g^{-1} (Figure 2g). Pore size distribution, derived from adsorption isotherms and simulated using the quenched solid density functional theory (QSDFT) model, showed a major pore size of 2.0 nm, aligning with the constructed crystalline structures (Figure 2g inset).

As anticipated, the polymerization of H_2P and An can also yield 3D H_2P-An-COF with an identical topology. Both Por-An-COFs (NiP-An-COF and H_2P-An-COF) have been systematically characterized (Figures 3 and S3−S14). HR-TEM analysis also unequivocally demonstrated the presence of a well-defined porous architecture in 3D H_2P-An-COF, with adjacent lattice fringes exhibiting a characteristic spacing of 2.4 nm (Figure S3). Nitrogen sorption measurements further revealed an exceptional BET surface area of 1922 m^2 g^{-1} and a predominant pore size of 2.2 nm (Figure S4), structural characteristics that closely resembled those of the 3D NiP-An-COF. The H_2P-An-COF exhibits a total pore volume of 1.01 cm^3 g^{-1}, which indicates the considerable mass transport capability in photoresponsive applications (Figure S5). The formation of imine linkages in 3D Por-An-COFs was validated by Fourier transform infrared (FT-IR) spectra, where the aldehydic C═O stretching vibration at 1697 cm^{-1} diminished, and the characteristic C═N stretching vibration at 1625 cm^{-1} appeared (Figures 3a and S6). Solid-state ^{13}C cross-polarization magic angle spinning nuclear magnetic resonance (CP-MAS NMR) spectra further confirmed the presence of imine bonds, as evidenced by the peak at 157 ppm (Figures 3b and S7). To quantitatively determine the ratio of the two linkers in the Por-An-COFs, we hydrolyzed the 3D H_2P-An-COF samples with DCl and measured their ^1H nuclear magnetic resonance (NMR) spectra. By integrating the resonance peak intensities, the H_2P and An linkers were found to be present in

https://doi.org/10.1021/jacs.5c09844
J. Am. Chem. Soc. 2025, 147, 30369−30379

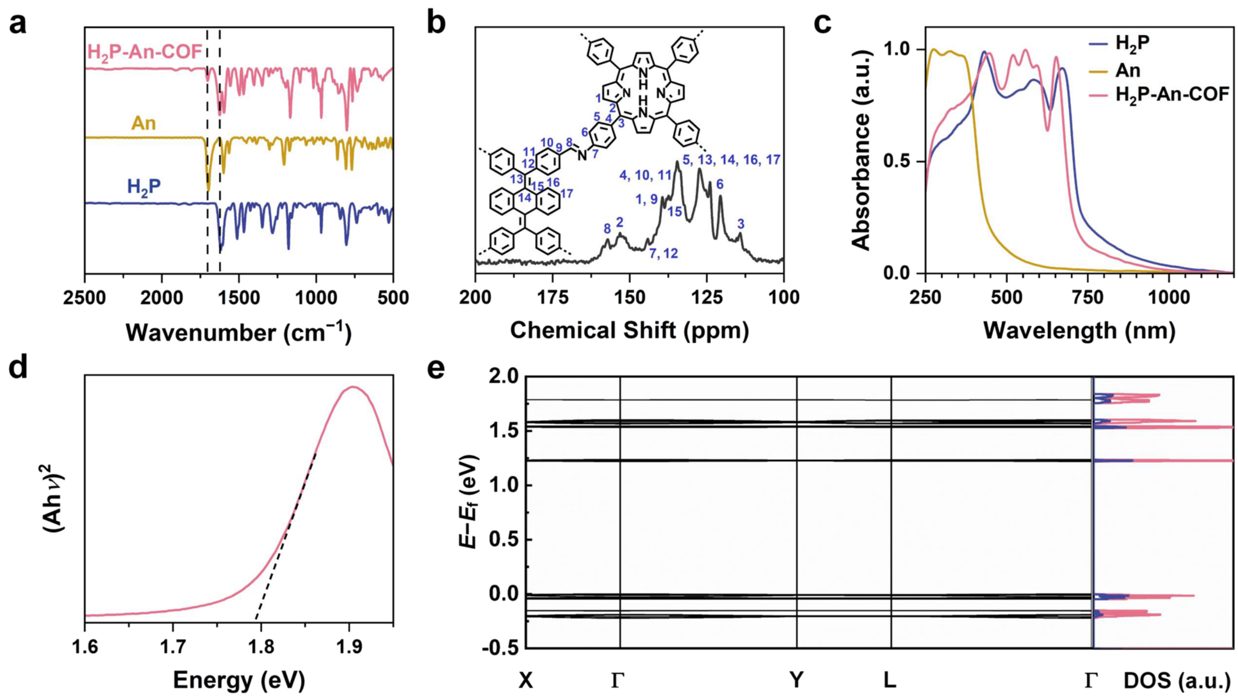


Figure 3. Chemical and electronic structures. (a) FT-IR spectra of H_2P (blue curve), An (yellow curve), and 3D H_2P-An-COF (red curve). The dashed line indicates the location of the characteristic peaks of C═O and C═N stretching vibrations. (b) Solid-state ^{13}C NMR spectrum and carbon signal assignment of 3D H_2P-An-COF with proposed peak assignments. (c) UV−vis−NIR spectra of H_2P (blue curve), An (yellow curve), and 3D H_2P-An-COF (red curve). (d) Tauc plot of 3D H_2P-An-COF. (e) Band structure and DOS analysis of 3D H_2P-An-COF.

a 1:1 ratio for 3D H_2P-An-COF (Figure S8). The integrations of protons quantitatively confirmed the lattice components of 3D H_2P-An-COF. Field-emission scanning electron microscopy (FE-SEM) images revealed that the 3D Por-An-COFs displayed a spherical, aggregated polymeric morphology (Figures S9 and S10). Energy-dispersive spectrometer (EDS) analysis confirmed the homogeneous distribution of carbon (C), nitrogen (N), and nickel (Ni) elements within the 3D Por-An-COFs (Figures S11 and S12). Thermogravimetric analysis (TGA) indicated that the 3D Por-An-COFs exhibited high thermal stability, remaining stable up to 400 °C under a nitrogen atmosphere (Figures S13 and S14).

To estimate the robustness of the covalently linked backbones, the 3D H_2P-An-COF was subjected to immersion in various solvents, including dichloromethane (DCM), tetrahydrofuran (THF), 1,4-dioxane, toluene, *N*,*N*-dimethylformamide (DMF), and an aqueous NaOH (12 M) solution for a period of 1 week. The residual weights of all 3D H_2P-An-COF samples exceeded 85% (Figure S15), with the preservation of their original PXRD patterns (Figure S16). The consistency of FT-IR spectroscopy further confirmed the maintenance of chemical structures (Figure S17). Additionally, the durability of the COF was corroborated by nitrogen sorption isotherms after the above-mentioned treatment (Figure S18). All these results demonstrate that the 3D H_2P-An-COF possesses outstanding stability against the corresponding common solvents.

The 3D H_2P-An-COF exhibits a broad absorption range from 250 to 1000 nm (Figure 3c, red curve), which is nearly the sum of the bands of H_2P and An monomers (Figure 3c, blue and yellow curves). The minor blue shift of absorption spectra of 3D H_2P-An-COF relative to H_2P indicates isolated locations of porphyrin units within the 3D framework instead of a close stacking state within H_2P powder. The optical band gap of 3D H_2P-An-COF, derived from the Tauc plot of the solid-state reflection spectra, is determined to be 1.79 eV (Figure 3d). This value is indicative of the potential for applications in photo energy conversion devices, as it suggests a narrow bandgap for efficient electron excitation via light irradiation.

The analysis of the band structure and density of states (DOS) (Figures 3e and S19) further illuminates the electronic properties of 3D H_2P-An-COF, demonstrating significant contribution from both the carbon p (Cp) and nitrogen p (Np) orbitals at the conduction band minimum (CBM) and the valence band maximum (VBM). Additionally, Mott−Schottky analysis confirms that 3D H_2P-An-COF demonstrates the characteristics of a *n*-type semiconductor (Figure S20). Density Functional Theory (DFT) calculations enable the investigation of the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) levels of the representative molecular unit extracted from the 3D H_2P-An-COF (Figure S21). The HOMO and LUMO levels for 3D H_2P-An-COF are determined to be −5.30 eV and −2.49 eV, respectively, at PBE0/def2-TZVP level. Due to the hybridization of node and linker units and extended π-conjugation within the 3D network, 3D H_2P-An-COF exhibits higher HOMO and lower LUMO levels compared to those of the H_2P monomer. These findings indicate that 3D H_2P-An-COF possesses semiconducting properties with a well-defined topotactic structure.

Photodegradation. The chemical warfare agent of sulfur mustard, known for its deleterious effects on human society, has become the subject of intensive research aimed at its effective degradation. This section focuses on the detoxification of sulfur mustard using 2-chloroethyl ethyl sulfide (CEES) as a proxy and employs 3D H_2P-An-COF as a heterocatalyst. The conversion of CEES to 1-chloro-2-ethylsulfinyl ethane

https://doi.org/10.1021/jacs.5c09844
J. Am. Chem. Soc. 2025, 147, 30369−30379

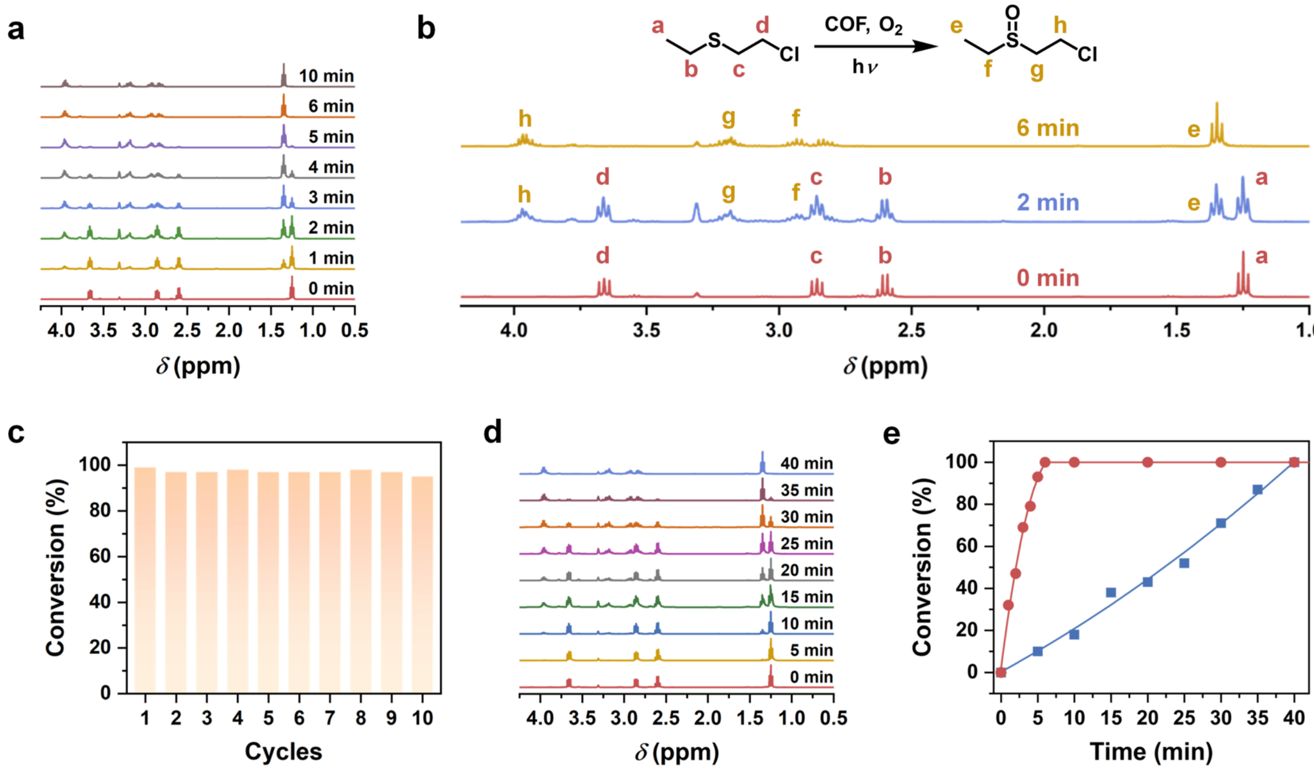


Figure 4. Photooxidation analysis of 3D H_2P-An-COF and comparison with 2D COF-366. (a) ^{1}H NMR spectra of CEES before and after photooxidation reaction by 3D H_2P-An-COF catalyst under O_2. (b) Conversion details of photooxidation by 3D H_2P-An-COF. (c) Recycling performance of 3D H_2P-An-COF toward CEES photooxidation. (d) ^{1}H NMR spectra before and after the photooxidation reaction of CEES by the 2D COF-366 catalyst in O_2. (e) Comparison of 3D H_2P-An-COF (red curve) with 2D COF-366 (blue curve) in conversion efficiency.

(CEESO) was monitored under conditions of white light irradiation, utilizing deuterated methanol as the solvent. In the presence of 3D H_2P-An-COF, CEES underwent complete degradation to CEESO within a remarkably short period of 6 min, with a half-life ($t_{1/2}$) of 2 min, indicative of exceptional catalytic efficiency (Figure 4a). The conversion was confirmed by the disappearance of ^{1}H NMR peaks at 1.25, 2.59, 2.86, and 3.66 ppm, and the concurrent emergence of peaks at 1.35, 2.83, 2.93, 3.18, and 3.95 ppm, which corresponded to CEES and CEESO, respectively (Figure 4b). As anticipated, the NiP-An-COF exhibited significantly lower photodegradation efficiency compared to H_2P-An-COF, which was most probably caused by the constrained generation of the reactive oxygen species (ROS) within the nickel atom incorporated porphyrin core (Figure S22). The crystalline and chemical structures of the 3D H_2P-An-COF were found to be substantially preserved following the photodegradation, as evidenced by the PXRD patterns and FT-IR spectra (Figure S23), indicating its excellent chemical stability. To assess the recyclability of the photocatalyst, we conducted ten consecutive cycles. Throughout these cycles, the complete degradation of CEES to CEESO was consistently achieved, with a conversion efficiency over 95%. This result substantiates the robustness and high catalytic performance of the 3D H_2P-An-COF (Figure 4c). Next, to elucidate the ROS involved in the process, electron paramagnetic resonance (EPR) spectroscopy was employed to confirm the types of radicals. When 2,2,6,6-tetramethyl-4-piperidone (TEMP) was used as a spin-trapping agent, characteristic 1:1:1 signals of (2,2,6,6-tetramethylpiperidin-1-yl)oxyl (TEMPO) were observed. This indicates that 3D H_2P-An-COF effectively generates singlet oxygen ($^1O_2$) under light irradiation, promoting the degradation of CEES (Figure S24). Therefore, the proposed mechanism of the photodegradation procedure is as follows: Upon light irradiation, the singlet oxygen molecules are yielded from triplet oxygen through a catalytic process mediated by sterically exposed porphyrin units in 3D H_2P-An-COF and then proceed to oxidize CEES into the nontoxic CEESO (Figure S25).

For a better comparison, a series of 2D porphyrin-based COFs, namely COF-366,[43] COF-367,[44] and H_2P-DMTP-COF[45] were employed as control groups (Figures S26 and S27), which demonstrated a degradation efficiency within respective durations of 40, 35, and 30 min (Figures 4d and S28). The results reveal that the degradation tendency of the 3D H_2P-An-COF significantly outperforms that of other 2D COFs, manifesting a photocatalytic activity that is at least approximately 5-fold higher (Figures 4e and S28). This enhancement is attributed to the abundance of exposed active sites within the steric configuration of 3D H_2P-An-COF, which offers more opportunities for substrate binding and catalyst interaction, thereby promoting efficient reaction kinetics. Furthermore, all COFs demonstrated uniform and complete selectivity for the degradation of CEES, with no detectable byproducts, such as 1-chloro-2-ethylsulfonyl ethane ($CEESO_2$) and ethyl vinyl sulfoxide (EVSO). This selectivity is crucial for ensuring the safety and effectiveness of the degradation process, as it minimizes the formation of potentially harmful or unwanted side products. In addition, to unveil the correlation between the type of linkage and photoresponsive capabilities, we converted the imine linkage of 3D H_2P-An-COF to a quinoline-linked structure through a previously reported methodology.[46] The corresponding crystalline and

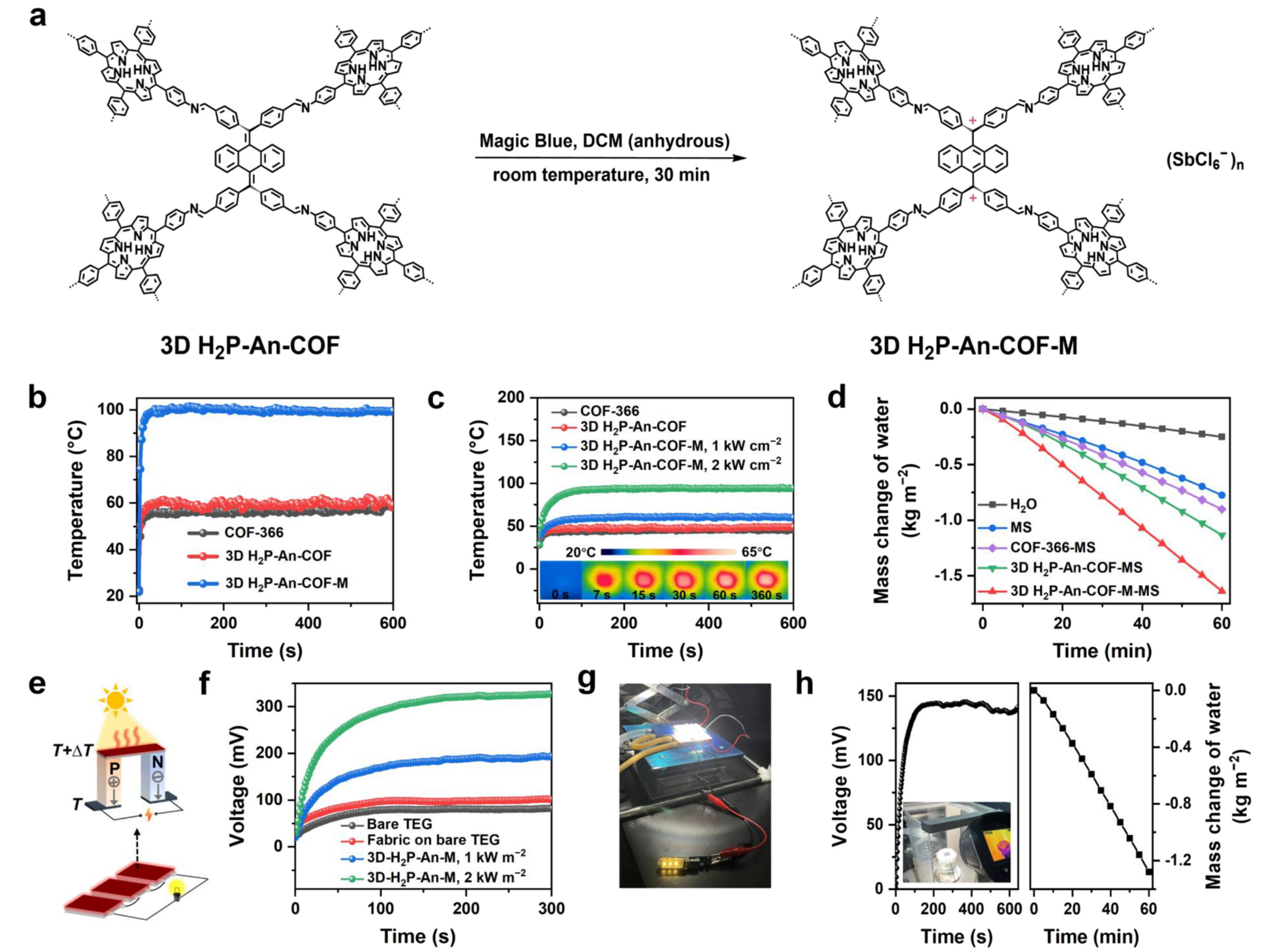


Figure 5. Photothermal properties and solar-thermal energy applications. (a) Synthesis of 3D H_2P-An-COF-M. (b) Temperature profiles of 3D H_2P-An-COF-M (blue curve), 3D H_2P-An-COF (red curve), and COF-366 (black curve) under irradiation of 808 nm laser at a power density of 0.25 W cm^{-2}. (c) Temperature profiles of 3D H_2P-An-COF (red curve), COF-366 (black curve), and 3D H_2P-An-COF-M (blue curve) under 1 kW m^{-2} simulated solar irradiation and of 3D H_2P-An-COF-M (green curve) under 2 kW m^{-2} simulated solar irradiation. Inset: IR thermal images of 3D H_2P-An-COF-M captured at different times during the heating process under 1 kW m^{-2} simulated solar irradiation. (d) Mass loss curves for bulk water (black curve), evaporation systems with blank MS evaporator (blue curve), COF-366-MS evaporator (purple curve), 3D H_2P-An-COF-MS evaporator (green curve), and 3D H_2P-An-COF-M-MS evaporator (red curve) for water during 1 h of simulated solar irradiation at 1 kW m^{-2}. (e) Schematic illustration of the power generation principle of the TEG and the configuration of the series-connected TEGs used to light up the LEDs. (f) Output voltages of bare TEG (black curve) and TEG with blank cellulose paper (red curve) on the hot side under 1 kW m^{-2} simulated solar irradiation, and the output voltages of 3D H_2P-An-COF-M-paper-based TEG under 1 kW m^{-2} (blue curve) and 2 kW m^{-2} (green curve) simulated solar irradiation. (g) Photograph of the thermoelectric device lighting up the LED lamp. (h) Output voltage (left) and mass loss curve (right) of the water-electricity cogeneration system under 1 kW m^{-2} simulated solar irradiation. Inset: photograph of a water-electricity cogeneration system.

chemical structures have been thoroughly characterized by PXRD (Figure S29) and FT-IR (Figure S30), thereby confirming the successful synthesis of quinoline-linked COF. As shown in Figure S31, the quinoline-linked COF demonstrated drastically mitigated photodegradation efficiency under identical conditions, highlighting the pivotal role of the imine linkage in H_2P-An-COF for achieving outstanding performance in this process. Considering the aforementioned structural features of COFs, which are synthesized through the polymerization of porphyrin units and predominantly aromatic carbon-based linkers, the superior performance of the 3D H_2P-An-COF, relative to the other three 2D COFs and quinoline-linked COF, highlights the potential benefits of 3D architectures and imine linkage in augmenting catalytic efficiency and selectivity for the detoxification of chemical warfare agents.

Photothermal Conversion. Given that the photothermal conversion performance highly depends on the light harvesting feature of COFs, exploring a pathway to further extend the absorption range of 3D H_2P-An-COF becomes crucial for achieving high light energy conversion efficiency. According to the literature, the absorption band of 11,11,12,12-tetraaryl-9,10-anthraquinodimethanes (AQDs) can be significantly red-shifted by generating robust dication states via one-stage two-electron oxidation,[47] which is potentially beneficial for photothermal conversion. As one typical AQD structure, the An units in 3D H_2P-An-COF can be fully oxidized by Magic Blue (MB), a process that has been meticulously characterized using solid-state ^{13}C CP-MAS NMR (Figures 5a and S32).

This transformation reaction produced a dark-orchid-colored material, designated as 3D H_2P-An-COF-M. EPR spectra were used to provide evidence of dicationic radicals. Under identical processing and measurement conditions, characteristic EPR signals emerged with increasing amounts of MB added, indicating a rise in dicationic radical concentration (Figure S33). The crystal and chemical structures of 3D H_2P-An-COF-M were meticulously characterized by PXRD, nitrogen sorption isotherms, and FT-IR spectra (Figures S34−S36). UV−vis−NIR diffuse reflectance spectra revealed that 3D H_2P-An-COF exhibited a weak absorption band spanning the range of 200−1000 nm. In sharp contrast, 3D H_2P-An-COF-M demonstrated substantially enhanced absorbance, with its absorption range extending to 2000 nm. This represents a significant red shift compared to the pristine 3D H_2P-An-COF (Figure S37). Additionally, the 3D H_2P-An-COF-M exhibited a reasonable electrical conductivity of 3.33×10^{-5} S m^{-1} at 30 °C and a thermal conductivity of 0.81 W m^{-1} K^{-1} at 50 °C, indicating its potential for photo-to-heat and heat-to-electrical conversion (Table S2).

Additionally, to elucidate the intrinsic correlation between the dimensionality of COFs and their propensity for photo-to-thermal conversion, 2D COF-366 was employed as a control alongside 3D H_2P-An-COF and 3D H_2P-An-COF-M for comparative analysis. Initially, the photothermal conversion capacity of 3D H_2P-An-COF-M was evaluated. Owing to its substantial absorption within the near-infrared (NIR) region (780−2000 nm), the temperature of 3D H_2P-An-COF-M rapidly increased from 22 °C to around 100 °C under irradiation by an 808 nm laser with a power density of 0.25 W cm^{-2}. In contrast, under the same conditions, the temperature elevation of bare 3D H_2P-An-COF and COF-366 was limited, reaching only around 60 and 57 °C, respectively (Figure 5b). The result can be attributed to their restricted NIR absorption capabilities. As the power density of the 808 nm laser was incremented to 0.45 and 0.70 W cm^{-2}, the temperature of 3D H_2P-An-COF-M correspondingly ascended to approximately 145 and 178 °C, respectively (Figure S38a). Furthermore, 3D H_2P-An-COF-M exhibited excellent photothermal stability, maintaining an equilibrium temperature of around 178 °C during each 10 min heating phase in a cycle of three heating and cooling processes with the 808 nm laser at 0.70 W cm^{-2} (Figure S38b). The superior photothermal performance of 3D H_2P-An-COF-M was also validated under simulated solar irradiation. Remarkably, 3D H_2P-An-COF-M demonstrated a swift temperature elevation, achieving equilibrium temperatures of approximately 61 and 94 °C at irradiation intensities of 1 and 2 kW m^{-2}, respectively. This performance surpassed that of 3D H_2P-An-COF, 2D COF-366, and 2D COF-366-M, which reached equilibrium temperatures of approximately 49, 45, and 56 °C, respectively, under 1 kW m^{-2} irradiation (Figures 5c and S39). More importantly, this result also exceeds that of the 3D NiP-An-COF and NiP-An-COF-M under the identical conditions, demonstrating that the incorporation of nickel atom into the porphyrin unit negatively impacts the photo-to-thermal conversion process in this system (Figure S40). Besides, the equilibrium temperature of quinoline-linked COFs was found to be substantially lower than those of imine-linked COFs in both the oxidized and nonoxidized states. This observation underscores that the imine linkage within H_2P-An-COF is crucial for reaching exceptional photoresponsive performance (Figure S41).

Throughout three cycles of light on and off, the 3D H_2P-An-COF-M maintained a consistent equilibrium temperature under both 1 and 2 kW m^{-2} irradiation, thereby evidencing outstanding photothermal stability upon solar exposure (Figure S42). Remarkably, the well-preserved original diffraction signals observed in the PXRD, along with the identical vibration peaks presenting in the FT-IR spectra, fully verify the stable nature of H_2P-An-COF-M architecture (Figures S43 and S44). The enhanced photothermal performance of 3D H_2P-An-COF-M, as compared to the pristine 3D H_2P-An-COF and 2D COF-366, can be attributed to three primary factors: (1) an extended NIR absorption spectrum, which facilitates increased capture of solar energy, (2) distinctive structural features, such as the presence of a dicationic state within a large conjugated framework, which promotes efficient energy conversion, and (3) light can penetrate deeper due to the steric open pathways of interconnected pore network in 3D COFs, which can effectively mitigate photon energy loss caused by surface scattering phenomena. A comparative analysis against key performance metrics of other COFs, metal−organic frameworks (MOFs), and nonporous materials reported in the literature reveals that the H_2P-An-COF-M exhibits outstanding photothermal conversion efficiency. This enhanced performance underscores the significance of the unique, well-defined steric lattice in boosting the photoresponsive capability (Table S3). The proposed mechanism for solar energy conversion is shown in Figure S45. Solar excitation promotes electrons in 3D H_2P-An-COF to an excited state (S_1), followed by energy dissipation to the lowest vibrational level of the ground state (S_0) via vibrational relaxations within S_1 and S_0 states and nonradiative transitions from S_1 to S_0. This process releases thermal energy, thereby facilitating solar-driven photothermal conversion of the 3D H_2P-An-COF. Upon oxidation by Magic Blue, the obtained 3D H_2P-An-COF-M exhibits a narrowed bandgap. This diminished bandgap, in conjunction with the π-conjugated configuration, synergistically enhances nonradiative transitions, which in turn augments the photothermal conversion efficiency of the material.

The broadband light absorption and the efficient conversion of this energy into photothermal form by 3D H_2P-An-COF-M have inspired extensive research into their prospective applications within the realms of solar thermal energy conversion and utilization. Notably, these applications include the solar-driven interfacial evaporation of water and the generation of thermoelectric power, both of which hold promise for addressing the global challenges of energy and water scarcity. In particular, the 3D H_2P-An-COF (in both the oxidized and nonoxidized states), with its all-organic composition, stands out from the conventional heavy metal-based photoresponsive materials, thereby mitigating the potential for environmental contamination. The highly porous and water-resistant nature of these COFs simplifies the separation and recovery processes, thereby further reducing their environmental impact during utilization.

Exploring 3D H_2P-An-COF-M as advanced photothermal materials, a solar-driven interfacial water evaporation system was established and characterized, as shown in Figure S46. The 3D H_2P-An-COF-M powders (20 mg) were dispersed and loaded into a melamine sponge substrate (2.0 cm × 2.0 cm × 0.9 cm) to fabricate a photothermal evaporator, yielding a composite named 3D H_2P-An-COF-M-MS. A bulk melamine sponge, coupled with a cellulose paper strip, was used to

https://doi.org/10.1021/jacs.5c09844
J. Am. Chem. Soc. 2025, 147, 30369−30379

provide support and a water supply. The 3D H_2P-An-COF-M-MS was embedded in an insulating expanded polyethylene (EPE) frame and secured at the top of the water container, exposing a 2.0 cm × 2.0 cm projected area for solar absorption and consequent evaporation. Under 1 kW m^{-2} simulated solar irradiation, the surface temperature of the 3D H_2P-An-COF-M-MS increased from 17 to 36 °C within 20 min intervals, subsequently stabilizing at around 38 °C. Compared to the dark state, the water evaporation rate was significantly accelerated. During the 1 h evaporation process, the 3D H_2P-An-COF-M-MS demonstrated excellent evaporation performance, with an evaporation rate of 1.64 kg m^{-2} h^{-1} and solar-to-vapor conversion efficiency of 93.4%. The evaporation rate was significantly higher than that observed for bulk water (0.25 kg m^{-2} h^{-1}) and the evaporator built from blank melamine sponge (0.78 kg m^{-2} h^{-1}). Notably, the 3D H_2P-An-COF-M-MS also exhibits robust performance with evaporation rates of 1.57 and 1.54 kg m^{-2} h^{-1} when operating in saline water (3.5% NaCl) and dye-containing saline water (3.5% NaCl + 150 mg/L methyl blue) under 1 kW m^{-2} irradiation, respectively (Figure S47). These rates are slightly lower than the rate in pure water, primarily due to the vapor pressure depression caused by dissolved salts and dyes.

Meanwhile, evaporators prepared with 3D H_2P-An-COF and COF-366 as photothermal materials, following the identical methodology as employed for 3D H_2P-An-COF-M-MS, yielded lower evaporation rates of 1.13 and 0.90 kg m^{-2} h^{-1}, respectively (Figure 5d). This result is attributed to the attenuated solar absorption and inferior photothermal conversion ability of these control materials. Additionally, the absence of hydrophilic moieties (Figure S48), such as dicationic units present in 3D H_2P-An-COF-M-MS, serves as a pivotal factor contributing to the diminished performance of these comparative systems. We anticipate that the continuous optimization of the oxidative approach, in tandem with enhancements to the evaporator's optical architecture and thermal management systems, will contribute to further improvements in solar-to-vapor conversion efficiency in the future.

To fully harness the thermal energy derived from solar photothermal conversion, 3D H_2P-An-COF-M was further implemented into a photothermo-electric energy conversion system (Figure 5e). The temperature difference between two dissimilar electrical conductors or semiconductors of this system induces an electric potential based on the Seebeck effect. 3D H_2P-An-COF-M was uniformly distributed on a cellulose paper substrate (3.0 cm × 3.0 cm), denoted as "3D H_2P-An-COF-M-paper", which functioned as the solar absorber and was coated on the hot side of a thermoelectric generator (TEG). The cold side of the TEG maintained thermal contact with a water-circulating cooling unit. Upon exposure to solar irradiation, the 3D H_2P-An-COF-M-paper heated up rapidly, creating a pronounced temperature gradient across the two sides of the TEG, which in turn led to an elevation in the voltage output. As depicted in Figures 5f and S49, the voltage output of the 3D H_2P-An-COF-M-paper based TEG increased with the rising temperature of the hot side, achieving a maximum voltage of 195 mV within 5 min under 1 kW m^{-2} irradiation. This performance demonstrated substantial net enhancement of 113 mV and 93 mV compared to the bare TEG (82 mV) and TEG with a blank cellulose paper on the hot side (102 mV), respectively, under identical conditions. Upon increasing the irradiation intensity to 2 kW m^{-2}, the output voltage of the 3D H_2P-An-COF-M-paper based TEG escalated to a maximum value of 328 mV. The cycle stability and durability of the TEG were further validated through three cycles of irradiation on and off. During each cycle of irradiation, which lasted approximately 10 min, the device maintained a stable and consistent peak output voltage, thereby demonstrating the excellent reliability of the TEG (Figure S50). The practical application potential of the TEG was further explored by connecting three 3D H_2P-An-COF-M-paper based TEGs in series, yielding a total output voltage of 0.59 V under 1 kW m^{-2} simulated solar irradiation. Through further voltage amplification using a boost module, the output voltage was amplified to 5 V, providing sufficient energy to light up the miniature commercial LED lamps (Figures 5g and S51, Video 1). The exceptional photothermo-electric energy conversion is most probably attributed to the robust light harvesting capacity across a broad absorption spectrum and the unique dicationic conjugated network within 3D H_2P-An-COF-M, which facilitates efficient carrier and heat transport. Notably, the 3D 2-fold lvt-b topotactic COFs stand out from both conventional carbon-based materials (e.g., carbon black, graphene oxide) and conjugated polymers (e.g., polypyrrole, polyaniline), which lack porous nanostructures with high surface area, as well as from 2D photoresponsive COFs. The 3D architecture ensures the rapid mass and energy transport, and reduces the loss of solar energy caused by the on-surface scattering, which together contribute to the high-performance of this smart photothermal system.

Ultimately, an integrated water-electricity cogeneration system was developed by combining interfacial water evaporation with thermoelectric power generation. This dual-purpose system is capable of producing water vapor under solar irradiation and simultaneously harnessing waste heat to generate electricity. The inset of Figure 5h illustrates the experimental setup of the integrated system. The 3D H_2P-An-COF-M-paper (comprising 20 mg of 3D H_2P-An-COF-M distributed over a 2.0 cm × 2.0 cm cellulose paper, Figure S52) was positioned on the hot side of the TEG, with an edge-attached cellulose paper strip serving as a water channel, while the cold side of TEG was in contact with water, supported by an insulating EPE frame at the top of the water container. The 3D H_2P-An-COF-M-paper worked as a planar evaporator, generating thermal energy during solar-driven water evaporation to drive power generation in the TEG via the established temperature difference. Consequently, the water-electricity cogeneration system achieved both efficient water evaporation with a rate of 1.28 kg m^{-2} h^{-1} and a voltage output of around 140 mV under 1 kW m^{-2} simulated solar irradiation (Figure 5h). The successful implementation of the 3D H_2P-An-COF-M in solar-driven interfacial water evaporation, electricity power generation, and water-electricity cogeneration underscores its promising potential for applications in solar-thermal energy conversion. It is worth noting that the distinctive chemical structure and molecular configuration of the AQDs functional unit are also responsible for the aforementioned characteristics. The pronounced steric repulsion between the hydrogen atoms of benzene rings and the anthracene cores dictates the nonplanar, twisted configurations of the An unit, which contribute to the formation of a 3D stable skeleton of Por-An-COF with sterically exposed active sites. Coupled with a 2-fold interpenetrating lattice, the fabrication of 3D COFs can be efficiently achieved without the need for intricate purification and activation processes, which are commonly employed to prevent the structural collapse. Furthermore, the An unit, composed entirely of sp^2 carbons, ensures the generation of 3D fully conjugated networks, which are potentially available to be decorated with well-known oxidants. Therefore, their electronic structures can be further well-tuned for specific semiconducting applications.

■ CONCLUSIONS

In summary, we have successfully explored a novel class of 3D porphyrin-based COFs with a lvt-b topotactic lattice. The 2-fold interpenetrated network exhibited high crystallinity, steric nanochannels, and extensive conjugated backbones. The periodic incorporation of porphyrin and AQD units endows these 3D architectures with exceptional photoresponsive features, making them promising candidates for solar-driven conversion of mass and energy. These COFs demonstrated remarkable photodegradation and photothermal conversion performance, which can be attributed to their outstanding light-harvesting nature and unique 3D configuration, enabling efficient interaction with guest molecules and rapid transport of heat and charge carriers.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c09844.

Additional experimental details for the synthesis of 3D NiP-An-COF, 3D H_2P-An-COF, and 2D COFs structure simulations, related characterizations (PXRD patterns, SEM images, TEM images, TGA, FT-IR spectra, gas sorption isotherms, and ^{13}C CP-MAS NMR spectra), additional photodegradation data, and photothermal conversion data (PDF)

Thermoelectric device lighting up the LED lamp (MP4)

■ AUTHOR INFORMATION

Corresponding Authors

Di Li − *Key Laboratory of Automobile Materials, School of Materials Science and Engineering, Jilin University, Changchun 130012, P. R. China*; Email: dili@jlu.edu.cn

Donglin Jiang − *Department of Chemistry, Faculty of Science, National University of Singapore, Singapore 117543, Singapore*; orcid.org/0000-0002-3785-1330; Email: chmjd@nus.edu.sg

Enquan Jin − *State Key Laboratory of Inorganic Synthesis and Preparative Chemistry, College of Chemistry and International Center of Future Science, Jilin University, Changchun 130012, P. R. China*; orcid.org/0009-0008-5043-5543; Email: enquanjin@jlu.edu.cn

Authors

Jiaqi Guo − *State Key Laboratory of Inorganic Synthesis and Preparative Chemistry, College of Chemistry and International Center of Future Science, Jilin University, Changchun 130012, P. R. China*

Yunyu Guo − *State Key Laboratory of Inorganic Synthesis and Preparative Chemistry, College of Chemistry and International Center of Future Science, Jilin University, Changchun 130012, P. R. China*

Mingxuan Zhang − *College of Chemistry and Molecular Engineering, Beijing National Laboratory for Molecular Sciences, Peking University, Beijing 100871, P. R. China*

Hongde Yu − *Faculty of Chemistry and Food Chemistry, TU Dresden, 01069 Dresden, Germany*; orcid.org/0000-0002-2576-271X

Haohao Bi − *Key Laboratory of Automobile Materials, School of Materials Science and Engineering, Jilin University, Changchun 130012, P. R. China*

Jingyi Feng − *State Key Laboratory of Inorganic Synthesis and Preparative Chemistry, College of Chemistry and International Center of Future Science, Jilin University, Changchun 130012, P. R. China*

Jiahuan Wang − *State Key Laboratory of Inorganic Synthesis and Preparative Chemistry, College of Chemistry and International Center of Future Science, Jilin University, Changchun 130012, P. R. China*

Keyu Geng − *Department of Chemistry, Faculty of Science, National University of Singapore, Singapore 117543, Singapore*

Thomas Heine − *Faculty of Chemistry and Food Chemistry, TU Dresden, 01069 Dresden, Germany; Helmholtz-Zentrum Dresden-Rossendorf, Centrum for Advanced Systems Understanding, CASUS, 02826 Görlitz, Germany; Department of Chemistry, Yonsei University and IBS Center for Nanomedicine, Seoul 120-749, Republic of Korea*; orcid.org/0000-0003-2379-6251

Junliang Sun − *College of Chemistry and Molecular Engineering, Beijing National Laboratory for Molecular Sciences, Peking University, Beijing 100871, P. R. China*; orcid.org/0000-0003-4074-0962

Complete contact information is available at: https://pubs.acs.org/10.1021/jacs.5c09844

Author Contributions

○J.G. and Y.G. contributed equally to this work.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the National Key Research and Development Program of China (Grant 2024YFB3815700), the National Natural Science Foundation of China (Grant 22371087), and the 111 center (B17020). E.J. acknowledges support from the start-up grant of Jilin University. D.J. acknowledges Singapore Ministry of Education (MOE) Tier 2 grant MOE-T2EP10221-0006. D.L. acknowledges support from the Jilin Province Science and Technology Research Project (Grant YDZJ202501ZYTS292). We thank Dr. Zonglong Li for his constructive suggestions on analyzing the potential crystalline structures and Prof. Tianshuang Wang for his assistance on recording conductivity.

■ REFERENCES

(1) Wang, X.; Maeda, K.; Thomas, A.; Takanabe, K.; Xin, G.; Carlsson, J. M.; Domen, K.; Antonietti, M. A metal-free polymeric photocatalyst for hydrogen production from water under visible light. *Nat. Mater.* **2009**, *8*, 76−80.

(2) Gu, J.; Peng, Y.; Zhou, T.; Ma, J.; Pang, H.; Yamauchi, Y. Porphyrin-based framework materials for energy conversion. *Nano Res. Energy* **2022**, *1*, No. e9120009.

(3) Côté, A. P.; Benin, A. I.; Ockwig, N. W.; O'Keeffe, M.; Matzger, A. J.; Yaghi, O. M. Porous, crystalline, covalent organic frameworks. *Science* **2005**, *310*, 1166−1170.

(4) Jiang, D.; Tan, V. G. W.; Gong, Y.; Shao, H.; Mu, X.; Luo, Z.; He, S. Semiconducting covalent organic frameworks. *Chem. Rev.* **2025**, *125*, 6203−6308.

(5) Moisanu, C. M.; Eckvahl, H. J.; Stern, C. L.; Wasielewski, M. R.; Dichtel, W. R. A paired-ion framework composed of vanadyl porphyrin molecular qubits extends spin coherence times. *J. Am. Chem. Soc.* **2024**, *146*, 28088−28094.

(6) Wang, X.; Chen, L.; Chong, S. Y.; Little, M. A.; Wu, Y.; Zhu, W.-H.; Clowes, R.; Yan, Y.; Zwijnenburg, M. A.; Sprick, R. S.; et al. Sulfone-containing covalent organic frameworks for photocatalytic hydrogen evolution from water. *Nat. Chem.* **2018**, *10*, 1180−1189.

(7) Ma, T.; Kapustin, E. A.; Yin, S. X.; Liang, L.; Zhou, Z.; Niu, J.; Li, L.-H.; Wang, Y.; Su, J.; Li, J.; et al. Single-crystal x-ray diffraction structures of covalent organic frameworks. *Science* **2018**, *361*, 48−52.

(8) Mahato, A. K.; Paul, S.; Banerjee, R. Synthesis innovations for crystallizing covalent organic framework thin films on biological and non-biological substrates. *Chem. Soc. Rev.* **2025**, *54*, 3578−3598.

(9) Wang, Z.; Zhang, Z.; Qi, H.; Ortega-Guerrero, A.; Wang, L.; Xu, K.; Wang, M.; Park, S. W.; Hennersdorf, F.; Dianat, A.; et al. On-water surface synthesis of charged two-dimensional polymer single crystals via the irreversible Katritzky reaction. *Nat. Synth.* **2022**, *1*, 69−76.

(10) Zhao, Y.; Dai, W.; Peng, Y.; Niu, Z.; Sun, Q.; Shan, C.; Yang, H.; Verma, G.; Wojtas, L.; Yuan, D.; et al. A corrole-based covalent organic framework featuring desymmetrized topology. *Angew. Chem., Int. Ed.* **2020**, *59*, 4354−4359.

(11) Han, X.; Yuan, C.; Hou, B.; Liu, L.; Li, H.; Liu, Y.; Cui, Y. Chiral covalent organic frameworks: design, synthesis and property. *Chem. Soc. Rev.* **2020**, *49*, 6248−6272.

(12) Mishra, B.; Alam, A.; Chakraborty, A.; Kumbhakar, B.; Ghosh, S.; Pachfule, P.; Thomas, A. Covalent organic frameworks for photocatalysis. *Adv. Mater.* **2024**, No. 2413118.

(13) Chen, F.; Zheng, H.; Yusran, Y.; Li, H.; Qiu, S.; Fang, Q. Exploring high-connectivity three-dimensional covalent organic frameworks: topologies, structures, and emerging applications. *Chem. Soc. Rev.* **2025**, *54*, 484−514.

(14) Keller, N.; Bein, T. Optoelectronic processes in covalent organic frameworks. *Chem. Soc. Rev.* **2021**, *50*, 1813−1845.

(15) Martín-Illán, J. Á.; Sierra, L.; Guillem-Navajas, A.; Suárez, J. A.; Royuela, S.; Rodríguez-San-Miguel, D.; Maspoch, D.; Ocón, P.; Zamora, F. β-ketoenamine-linked covalent organic frameworks synthesized via gel-to-gel monomer exchange reaction: from aerogel monoliths to electrodes for supercapacitors. *Adv. Funct. Mater.* **2024**, *34*, No. 2403567.

(16) He, T.; Zhao, Z.; Liu, R.; Liu, X.; Ni, B.; Wei, Y.; Wu, Y.; Yuan, W.; Peng, H.; Jiang, Z.; et al. Porphyrin-based covalent organic frameworks anchoring Au single atoms for photocatalytic nitrogen fixation. *J. Am. Chem. Soc.* **2023**, *145*, 6057−6066.

(17) Wang, Y.-R.; Ding, H.-M.; Sun, S.-N.; Shi, J.-W.; Yang, Y.-L.; Li, Q.; Chen, Y.; Li, S.-L.; Lan, Y.-Q. Light, heat and electricity integrated energy conversion system: photothermal-assisted co-electrolysis of CO_2 and methanol. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202212162.

(18) Jin, F.; Lin, E.; Wang, T.; Yan, D.; Yang, Y.; Chen, Y.; Cheng, P.; Zhang, Z. Rationally fabricating 3D porphyrinic covalent organic frameworks with scu topology as highly efficient photocatalysts. *Chem.* **2022**, *8*, 3064−3080.

(19) Kim, Y. H.; Jeon, J.-P.; Kim, Y.; Noh, H.-J.; Seo, J.-M.; Kim, J.; Lee, G.; Baek, J.-B. Cobalt-porphyrin-based covalent organic frameworks with donor-acceptor units as photocatalysts for carbon dioxide reduction. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202307991.

(20) Liu, Y.; Howarth, A. J.; Hupp, J. T.; Farha, O. K. Selective photooxidation of a mustard-gas simulant catalyzed by a porphyrinic metal−organic framework. *Angew. Chem., Int. Ed.* **2015**, *54*, 9001−9005.

(21) Gan, S.-X.; Jia, C.; Qi, Q.-Y.; Zhao, X. A facile and scalable synthetic method for covalent organic nanosheets: ultrasonic polycondensation and photocatalytic degradation of organic pollutants. *Chem. Sci.* **2022**, *13*, 1009−1015.

(22) Dai, C.; Liu, B. Conjugated polymers for visible-light-driven photocatalysis. *Energy Environ. Sci.* **2020**, *13*, 24−52.

(23) Chen, X.; Addicoat, M.; Jin, E.; Zhai, L.; Xu, H.; Huang, N.; Guo, Z.; Liu, L.; Irle, S.; Jiang, D. Locking covalent organic frameworks with hydrogen bonds: general and remarkable effects on crystalline structure, physical properties, and photochemical activity. *J. Am. Chem. Soc.* **2015**, *137*, 3241−3247.

(24) Lin, G.; Ding, H.; Chen, R.; Peng, Z.; Wang, B.; Wang, C. 3D porphyrin-based covalent organic frameworks. *J. Am. Chem. Soc.* **2017**, *139*, 8705−8709.

(25) Wang, Q.-Y.; Liu, J.; Cao, M.; Hu, J.-H.; Pang, R.; Wang, S.; Asad, M.; Wei, Y.-L.; Zang, S.-Q. Aminal-linked porphyrinic covalent organic framework for rapid photocatalytic decontamination of mustard-gas simulant. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202207130.

(26) Lei, H.; Zhu, C.; Lin, L.; Han, X.; Ding, Y.; Song, J.; Tian, Y.; Zhu, G. Rational modulation of covalent organic frameworks heterogeneous catalyst: structural cationization effect on accelerating photocatalytic oxidation process. *Appl. Catal. B Environ. Energy* **2025**, *361*, No. 124654.

(27) Ji, W.; Wang, T.-X.; Ding, X.; Lei, S.; Han, B.-H. Porphyrin- and phthalocyanine-based porous organic polymers: from synthesis to application. *Coord. Chem. Rev.* **2021**, *439*, No. 213875.

(28) Nagai, A.; Chen, X.; Feng, X.; Ding, X.; Guo, Z.; Jiang, D. A squaraine-linked mesoporous covalent organic framework. *Angew. Chem., Int. Ed.* **2013**, *52*, 3770−3774.

(29) Zhang, Z.; Meng, F.; Chi, X.; Jiao, Y.; Xue, B.; Li, X.; Zhang, F. Diketopyrrolopyrrole-activated dynamic condensation approach to narrow-band gap vinylene-linked covalent organic frameworks. *Angew. Chem., Int. Ed.* **2025**, *64*, No. e202417805.

(30) Lin, Z.; Zhong, Y.-H.; Zhong, L.; Ye, X.; Chung, L.-H.; Hu, X.; Xu, Z.; Yu, L.; He, J. Minimalist design for solar energy conversion: revamping the π-grid of an organic framework into open-shell superabsorbers. *JACS Au* **2023**, *3*, 1711−1722.

(31) Ding, L.-G.; Wang, S.; Yao, B.-J.; Wu, W.-X.; Kan, J.-L.; Liu, Y.; Wu, J.; Dong, Y.-B. Covalent organic framework based multifunctional self-sanitizing face masks. *J. Mater. Chem. A* **2022**, *10*, 3346−3358.

(32) Mi, Z.; Yang, P.; Wang, R.; Unruangsri, J.; Yang, W.; Wang, C.; Guo, J. Stable radical cation-containing covalent organic frameworks exhibiting remarkable structure-enhanced photothermal conversion. *J. Am. Chem. Soc.* **2019**, *141*, 14433−14442.

(33) Tang, X.; Chen, Z.; Xu, Q.; Su, Y.; Xu, H.; Horike, S.; Zhang, H.; Li, Y.; Gu, C. Design of photothermal covalent organic frameworks by radical immobilization. *CCS Chem.* **2022**, *4*, 2842−2853.

(34) Zhu, Y.; Shao, P.; Hu, L.; Sun, C.; Li, J.; Feng, X.; Wang, B. Construction of interlayer conjugated links in 2D covalent organic frameworks via topological polymerization. *J. Am. Chem. Soc.* **2021**, *143*, 7897−7902.

(35) Ma, T.-R.; Ge, F.; Ke, S.-W.; Lv, S.; Yang, Z.-M.; Zhou, X.-C.; Liu, C.; Wu, X.-J.; Yuan, S.; Zuo, J.-L. Accessible tetrathiafulvalene moieties in a 3D covalent organic framework for enhanced near-infrared photo-thermal conversion and photo-electrical response. *Small* **2024**, *20*, No. 2308013.

(36) Jin, H.-J.; Zhao, P.-C.; Qian, Y.; Xiao, J.-D.; Chao, Z.-S.; Jiang, H.-L. Metal−organic frameworks for organic transformations by photocatalysis and photothermal catalysis. *Chem. Soc. Rev.* **2024**, *53*, 9378−9418.

(37) Zhao, L.; Liu, Y.; Xing, R.; Yan, X. Supramolecular photothermal effects: a promising mechanism for efficient thermal conversion. *Angew. Chem., Int. Ed.* **2020**, *59*, 3793−3801.

(38) Wen, F.; Xu, K.; Feng, Y.; Huang, N. Two-dimensional covalent organic frameworks with pentagonal pores. *J. Am. Chem. Soc.* **2024**, *146*, 19680−19685.

(39) Chen, R.; Shi, J.-L.; Ma, Y.; Lin, G.; Lang, X.; Wang, C. Designed synthesis of a 2D porphyrin-based sp^2 carbon-conjugated covalent organic framework for heterogeneous photocatalysis. *Angew. Chem., Int. Ed.* **2019**, *58*, 6430−6434.

(40) Ding, J.; Guan, X.; Lv, J.; Chen, X.; Zhang, Y.; Li, H.; Zhang, D.; Qiu, S.; Jiang, H.-L.; Fang, Q. Three-dimensional covalent organic frameworks with ultra-large pores for highly efficient photocatalysis. *J. Am. Chem. Soc.* **2023**, *145*, 3248−3254.

(41) Zheng, S.; Fu, Y.; Bi, S.; Yang, X.; Xu, X.; Li, X.; Xu, Q.; Zeng, G. Three-dimensional covalent organic framework with dense lithiophilic sites as protective layer to enable high-performance lithium metal battery. *Angew. Chem., Int. Ed.* **2025**, *64*, No. e202417973.

(42) Wan, W.; Sun, J.; Su, J.; Hovmöller, S.; Zou, X. Three-dimensional rotation electron diffraction: software RED for automated data collection and data processing. *J. Appl. Crystallogr.* **2013**, *46*, 1863−1873.

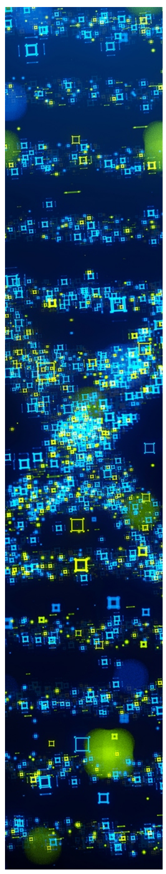
(43) Wan, S.; Gándara, F.; Asano, A.; Furukawa, H.; Saeki, A.; Dey, S.-K.; Liao, L.; Ambrogio, M. W.; Botros, Y. Y.; Duan, X.; et al. Covalent organic frameworks with high charge carrier mobility. *Chem. Mater.* **2011**, *23*, 4094−4097.

(44) Liu, W.; Li, X.; Wang, C.; Pan, H.; Liu, W.; Wang, K.; Zeng, Q.; Wang, R.; Jiang, J. A scalable general synthetic approach toward ultrathin imine-linked two-dimensional covalent organic framework nanosheets for photocatalytic CO_2 reduction. *J. Am. Chem. Soc.* **2019**, *141*, 17431−17440.

(45) Ye, X.; Liu, R.; Mu, X.; Tao, S.; Yang, H.; Gao, X. J.; Yang, S.-W.; Jiang, D. Superacid *in situ* protected synthesis of covalent organic frameworks. *J. Am. Chem. Soc.* **2025**, *147*, 6942−6957.

(46) Li, X.; Zhang, C.; Cai, S.; Lei, X.; Altoe, V.; Hong, F.; Urban, J. J.; Ciston, J.; Chan, E. M.; Liu, Y. Facile transformation of imine covalent organic frameworks into ultrastable crystalline porous aromatic frameworks. *Nat. Commun.* **2018**, *9*, 2998.

(47) Ishigaki, Y.; Harimoto, T.; Sugawara, K.; Suzuki, T. Hysteretic three-state redox interconversion among zigzag bisquinodimethanes with non-fused benzene rings and twisted tetra-/dications with [5]/[3] acenes exhibiting near-infrared absorptions. *J. Am. Chem. Soc.* **2021**, *143*, 3306−3311.



https://doi.org/10.1021/jacs.5c09844
J. Am. Chem. Soc. 2025, 147, 30369−30379